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Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 15

TITLE: TITLE : Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

Theory

In Java, **finally**, **throw**, and **throws** are important features used for effective exception handling.

- **finally block:** The finally block is executed after the try and catch blocks, whether an exception occurs or not. It is mainly used for **resource cleanup**, such as closing files, database connections, or other resources.
- **throw keyword:** The throw keyword is used to **explicitly create and generate an exception** when a specific condition occurs. For example, `throw new Exception("Invalid ATM PIN");` generates an exception when the entered PIN is incorrect.
- **throws keyword:** The throws keyword is used in a method declaration to **inform the caller that the method may generate an exception**. It passes the responsibility of handling that exception to the calling method.

Together, these mechanisms provide a **structured and reliable way to handle errors** in Java. They help prevent unexpected program termination, make programs easier to maintain, and improve overall application stability.

Exercises:

1. Create a Login program that throws an exception for invalid password and uses finally block.

Users > kingsushant > Desktop > JAVA > Login.java > Login

```
1  import java.io.*;
2
3  public class Login {
4
5      static void checkPassword(String password) throws Exception {
6          if (!password.equals("1234")) {
7              throw new Exception(message: "Invalid password");
8          }
9
10         System.out.println(x: "Login successful.");
11     }
12
13     public static void main(String[] args) {
14         try {
15             checkPassword(password: "123");
16
17         } catch (Exception e) {
18             System.out.println("Error: " + e.getMessage());
19
20         } finally {
21             System.out.println(x: "Login process completed.");
22         }
23     }
24 }
```

Output:

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
kingsushant@Sushant ~ % /usr/bin/env /opt/homebrew/Cellar/openjdk/26.0.1/libexec/openjdk.jdk/Contents/Home/bin/java
00gn/T/vscodesws_45438/jdt_ws/jdt.ls-java-project/bin Login
Error: Invalid password
Login process completed.
kingsushant@Sushant ~ %
```

2. Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

```
Users > kingsushant > Desktop > JAVA > J ATMPinVerification.java > ATMPinVerification > main(String[])
1  import java.io.*;
2
3  public class ATMPinVerification {
4
5      static void checkPin(int pin) throws Exception {
6          if (pin != 1234) {
7              throw new Exception(message: "Invalid ATM PIN");
8          }
9
10         System.out.println(x: "PIN verified successfully.");
11     }
12
13     public static void main(String[] args) {
14         try {
15             checkPin(pin: 1111);
16
17         } catch (Exception e) {
18             System.out.println("Error: " + e.getMessage());
19
20         } finally {
21             System.out.println(x: "PIN verification process completed.");
22         }
23     }
24 }
```

output:

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
kingsushant@Sushant ~ % /usr/bin/env /opt/homebrew/Cellar/openjdk/26.0.1/libexec/openjdk.jdk/Contents/Home/bin/java -
8n1fll8bjcq4jq40000gn/T/vscodesws_45438/jdt_ws/jdt.ls-java-project/bin ATMPinVerification
Error: Invalid ATM PIN
PIN verification process completed.
```

Conclusion:

The programs successfully demonstrate exception handling using finally, throw, and throws in Java. The Login program handles an invalid password, while the ATM PIN Verification program handles an invalid PIN. The throw keyword is used to explicitly generate an exception, and throws declares the exception in the method. The finally block executes after exception handling and displays the completion message. Thus, these programs demonstrate structured and reliable exception handling in Java.